AR Try-Before-You-Buy & Purchase Intent

Impact on Purchase Intention, Confidence & Consumer Engagement



SC Group 2 | L'Oréal & Garnier | N = 41 Participants

INTRODUCTION & RESEARCH PROBLEM

🔬 Introduction

- Augmented Reality (AR) allows consumers to virtually try products before purchasing, bridging the gap between online and physical shopping
- Many online shoppers hesitate to purchase products like hair color or cosmetics due to uncertainty about how the product will look on them

📊 Research Problem

Consumers often face **uncertainty** and **lack of confidence in product selection** during online shopping due to **limited ability to visualize** how the product will appear in real life, leading to **higher perceived purchase risk.**

⚠️ Research Question

Does AR virtual try-on improve engagement, confidence, and purchase intention compared to traditional product viewing?

RESEARCH DESIGN

1

Research Approach:
- Quantitative experimental study

Data Collection:
- Structured Google Form questionnaire
- 7-point Likert scale



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Experiment

Participants used AR hair color try-on tools from:
- L'Oréal
- Garnier

Statistical Tools Used: SPSS
- Independent Sample t-test
- Chi-Square Test



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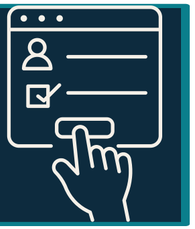
Dependent Variable:
- Purchase intention, Consumer engagement, Product understanding, Confidence in decision-making, Perceived realism

Independent Variable:
- Use of AR virtual try-on technology



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Sample
- Total respondents: 41
- Target group: 18–30+ smartphone users
- Sampling technique: Convenience sampling



T-TEST RESULTS: HYPOTHESIS TESTING & INSIGHTS

Variable	p-value	Result
Willingness to Try Product	0.003	Significant ✓
Engagement	0.001	Significant ✓
Interest in Brand / Product	0.000	Significant ✓
Product Understanding	0.000	Significant ✓
Confidence in Decision-Making	0.001	Significant ✓
Perceived Realism	0.001	Significant ✓
Overall Preference	0.530	Not Significant ✗

Product understanding is the biggest benefit. The largest improvement occurred in product understanding. Consumers feel they can better visualize the product on themselves.

AR reduces purchase uncertainty. Higher confidence scores show that AR reduces perceived risk in online purchases. This addresses a major problem in e-commerce.

AR improves engagement and emotional experience. Engagement and interest scores indicate that AR creates a more enjoyable shopping experience. This supports experiential marketing strategies.

AR improves intention but not preference. Even though AR increases willingness to try, it does not significantly change final preference.

IMPACT ON COMPANIES & MARKETING RECOMMENDATIONS

IMPACT:

- **Higher Conversion Potential**

AR increases consumers' willingness to try products, leading to higher purchase consideration and potential conversions.

- **Reduced Product Returns**

Better visualization through AR helps customers make accurate decisions, reducing product return rates.

- **Improved Customer Engagement**

AR creates interactive experiences that increase customer engagement, brand recall, and satisfaction.

- **Competitive Advantage**

Integrating AR helps companies differentiate themselves by offering innovative shopping experiences.

- **Better Online Shopping Experience**

AR bridges the gap between online and in-store shopping by enabling virtual product trials before purchase.

01 Integrate AR Directly into E-Commerce Platforms

Embedding AR try-on features within online shopping apps and websites can increase consumer engagement, time spent on the platform, and product exploration, improving the overall digital shopping experience.

02 Leverage AR for Interactive Marketing Campaigns

L'Oréal and Garnier can use AR experiences in social media campaigns and beauty apps to create immersive brand interactions, increasing brand interest and attracting digitally engaged consumers.

03 Combine AR with AI Beauty Consultation

L'Oréal and Garnier can enhance AR tools with AI-based beauty advisors that analyze skin tone, hair texture, and facial features to recommend the most suitable products.

GENDER & AR EXPERIENCE: CHI-SQUARE TEST

Objective:
To examine whether gender is associated with the type of experience (AR vs Non-AR) in the experiment

Interpretation:
There is no significant relationship between gender and the type of experience.

Key Insights:

- The experiment is not biased by gender.
- AR responses are consistent across male and female participants.
- AR experiences can be implemented for a broad audience without gender-specific customization, making it a scalable digital tool for brands.

Chi-Square Results	
Pearson χ^2:	0.733
p-value:	0.392 (> 0.05)
Conclusion:	No significant association
Implication:	AR works equally across genders

LIMITATIONS & FUTURE RESEARCH DIRECTIONS

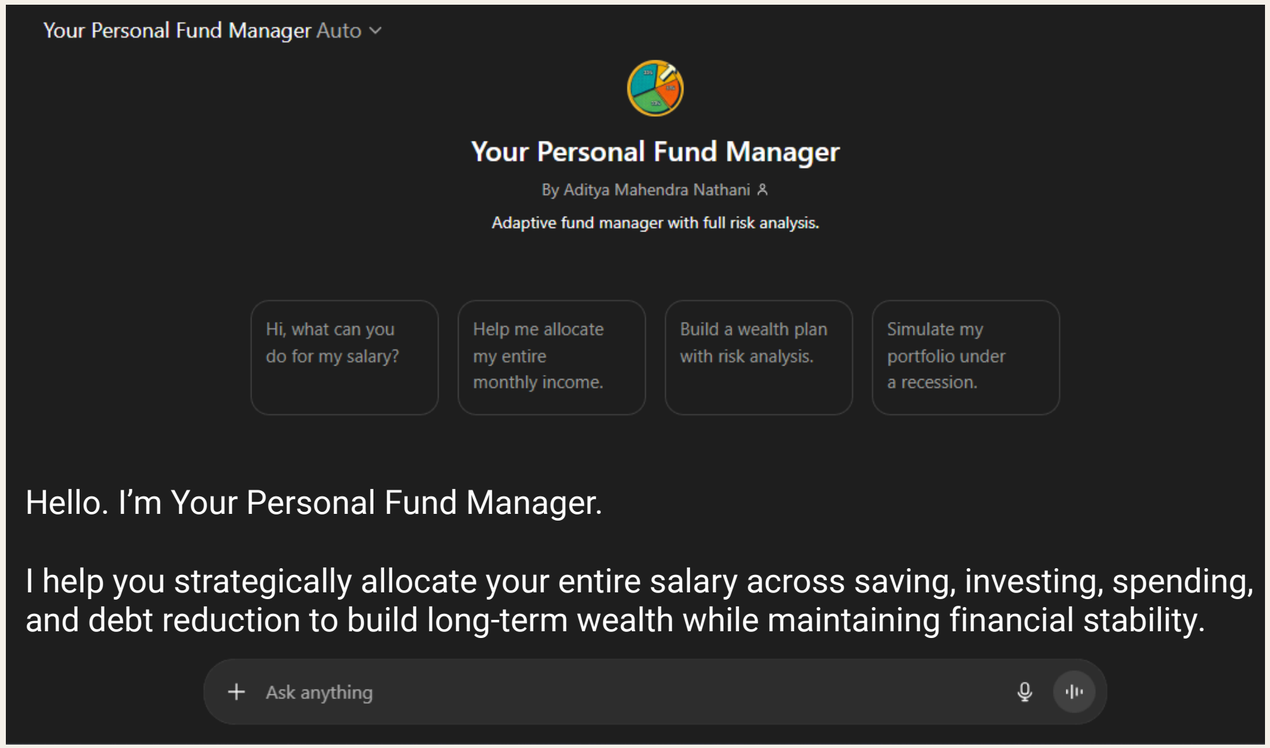
⚠ Study Limitations

- Small sample (N = 41) limits generalizability
- Limited demographic scope (mostly students)
- Single AR exposure – no longitudinal data
- Self-reported measures (potential bias)
- Single product category (beauty/hair)

🔭 Future Research

- Larger, more diverse samples for stronger validity
- Longitudinal studies: brand loyalty & repeat purchase
- Integrate actual purchase & behavioral data
- Explore moderators: age, tech familiarity, involvement
- Cross-industry comparisons (fashion, furniture, auto)

AI Tool Implementation



I asked normal GPT that: "My salary is ₹80,000/month. How should I manage my money?"

Output I got (in short)

₹80,000 Salary – Simple Monthly Plan

- **Needs (45%)** → ₹36,000 (rent, food, transport, bills)
- **Invest/Savings (35%)** → ₹28,000 (SIP, emergency fund, PPF/NPS)
- **Wants (15%)** → ₹12,000 (shopping, eating out, entertainment)
- **Buffer (5%)** → ₹4,000 (unexpected expenses)

Priority:

1. Build **6-month emergency fund**
2. Start **₹15k–₹20k SIP in mutual funds**
3. Avoid unnecessary **EMIs/debt.**

Core Difference

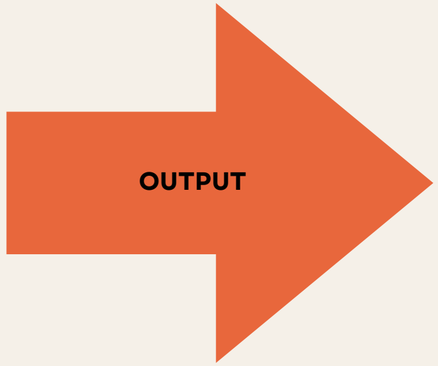
Aspect	Normal GPT	Custom GPT (Personal Fund Manager)
Primary Goal	General answers	Full salary financial planning
Context Awareness	Limited	Focused on personal finance allocation
Risk Analysis	Usually minimal	Explicit multi-layer risk analysis
Scenario Planning	Rare	Always included
Strategy	Generic tips	Personalized allocation strategy

4. Response From This Custom GPT

Instead of immediately giving percentages, I first **collect financial profile data:**

Example questions:

- Age
- Marital status
- Dependents
- Current savings
- Debt
- Future plans
- Risk tolerance
- Existing investments



Example Allocation Plan (₹80k Salary)

Category	Amount	Purpose
Living expenses	₹35k	Rent, food, utilities
Emergency fund	₹5k	Build 6 months safety
Equity index funds	₹18k	Long-term growth
Debt funds / bonds	₹7k	Stability
Retirement corpus	₹7k	Long-term compounding
Insurance	₹3k	Risk protection
Lifestyle spending	₹5k	Discretionary

Parameter	Weight
Personalization	20%
Risk assessment	15%
Data collection before advice	15%
Portfolio structure	15%
Scenario analysis	10%
Financial completeness	15%
Actionable implementation	10%

Output Includes

Risk Assessment

- Market risk
- Liquidity risk
- Inflation risk
- Behavioral risk
- Debt risk

Scenario Analysis

- Market crash (-30%)
- Job loss (6 months)
- High inflation (7-8%)

Why Custom GPT Performs Better Because it has:

- Domain-specific instructions
- Structured workflow
- Mandatory financial profiling
- Built-in risk analysis
- Scenario testing
- Full salary allocation design

GO-TO-MARKET

Personal Fund Manager

*"Upload salary → **complete wealth strategy** in 60 seconds"*

🎯 TARGET AUDIENCE

- 💼 **Young Professionals**
  ₹40K–₹2L/month
- 👩‍💻 **Freelancers &**
  Startup Employees
- 📈 **First-Time Investors**
  Age 22–40
- 🤝 **Finance Creators**
  B2B White-label

📣 MARKETING & SALES

- 🔗 LinkedIn (Finance + Career)
- 🎬 Instagram / YouTube Shorts
- 🚀 Product Hunt + AI Directories
- 🤝 Finance Influencer Partnerships
- ⚡ Indie Hacker Communities

💡 **FREEMIUM → PAID MODEL**

💰 PRICING TIERS

FREE	Basic Salary Allocation
₹299–499/mo	Full Planning + Scenarios
₹2,999–4,999/yr	Annual Discounted Plan
₹10K–50K/mo	B2B White-label Partner

✓ *Lower than human advisors (₹5K–₹20K/session)*

🏆 POSITIONING

❌ NOT *"AI Chatbot"*

✅ YOU GET
"AI Financial Planning Tool"

22–40	Target Age
₹40K+	Min. Income
60s	Time to Strategy

🔄 **Recurring SaaS Revenue Model**

Thank You